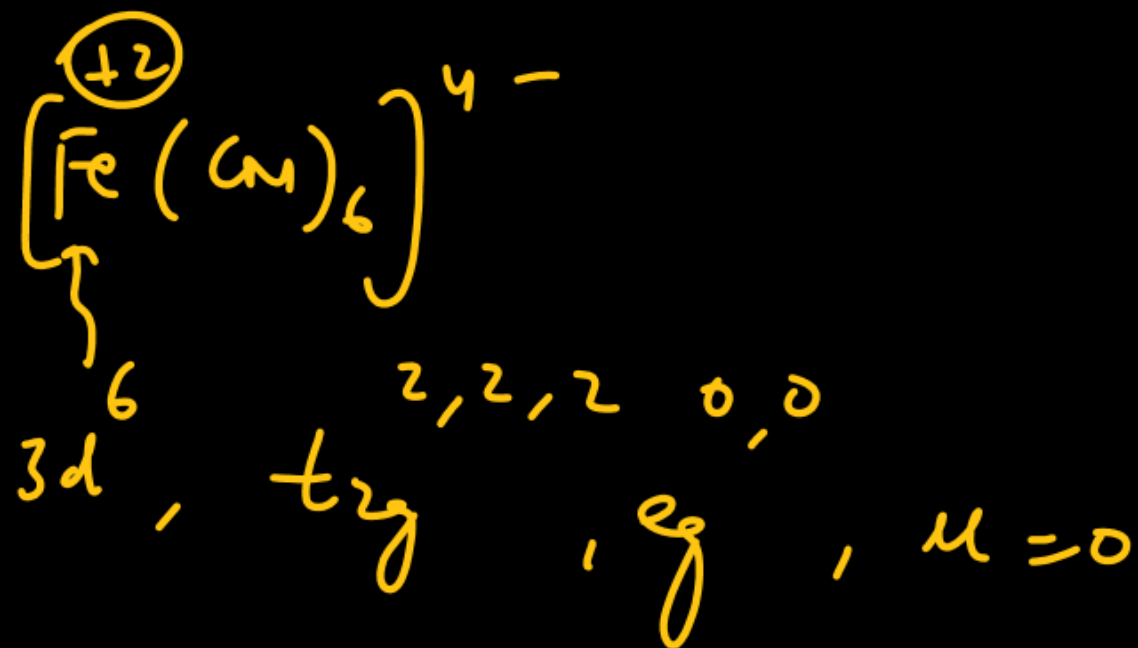
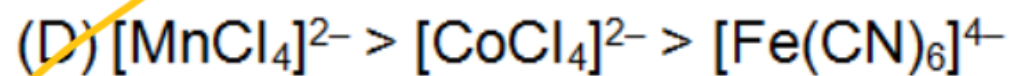
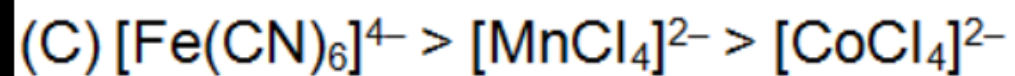
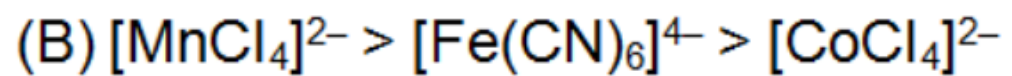
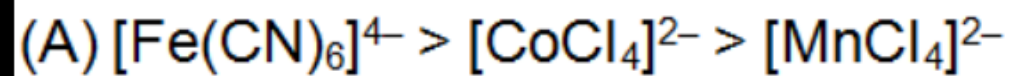
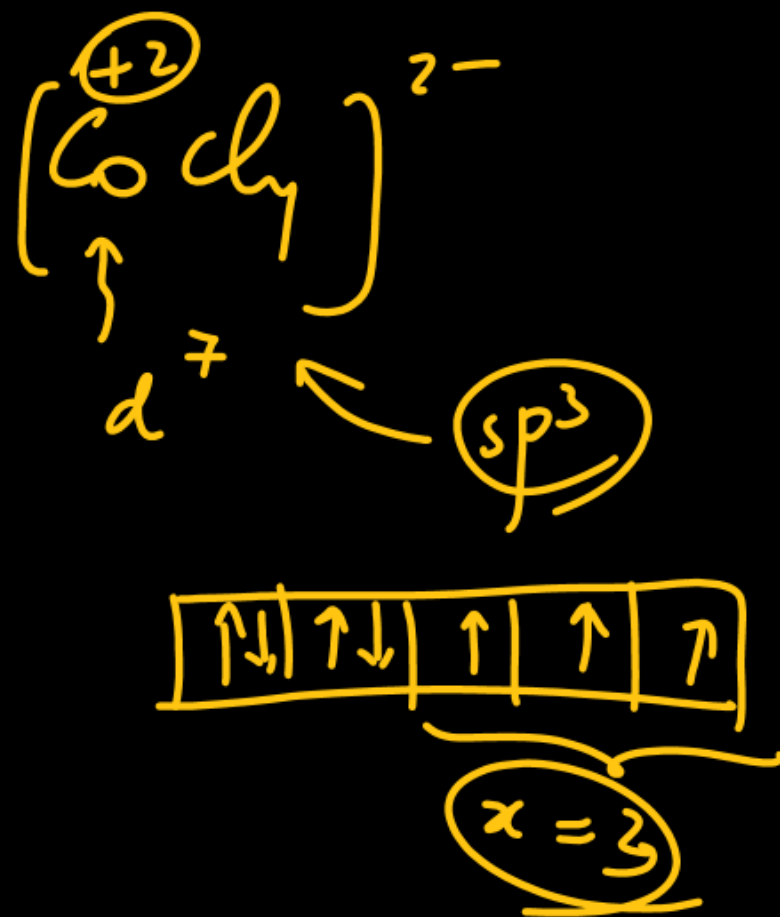


The correct order of magnetic moments (spin values in BM) among is :



$x = 0$

Ans : (D) ✓



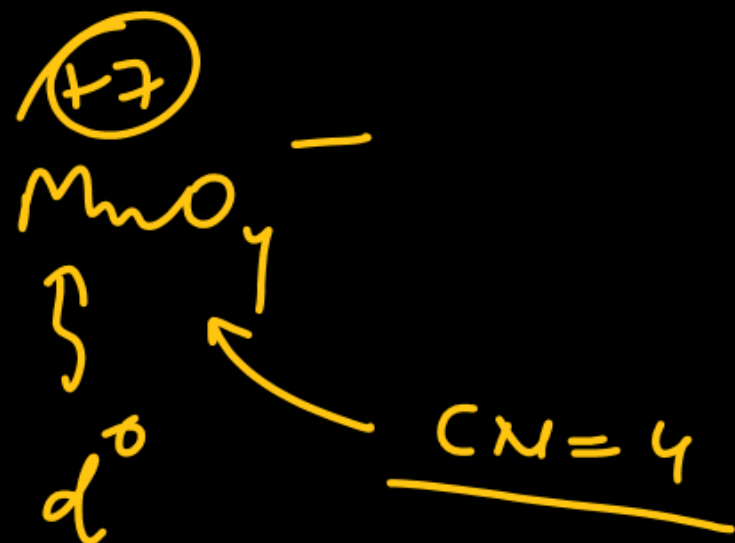
Set of d-orbitals which is used in hybridisation by central metal during formation of MnO_4^- ?

(A) $d_{x^2-y^2}, d_{z^2}, d_{xy}$

✓ (B) d_{xy}, d_{yz}, d_{xz}

(C) $d_{x^2-y^2}, d_{xy}, d_{xz}$

(D) $d_{x^2-y^2}, d_{z^2}, d_{xz}$



Ans: (B)

$d^3 s$ hybrid $\rightarrow d_{xy}, d_{yz}, d_{zx} + s$

($\text{TiCl}_4, \text{CrO}_4^{2-}, \text{CrO}_2\text{Cl}_2, \text{VO}_4^{3-}$
 MnO_4^- etc)

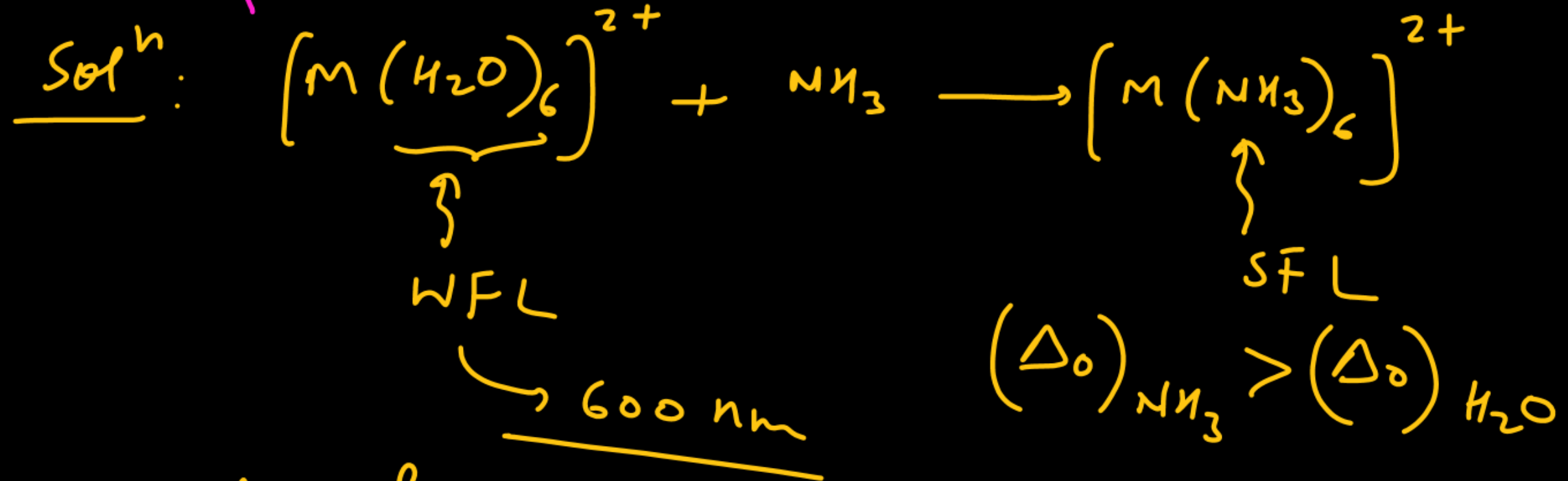
A complex $[M(H_2O)_6]^{2+}$ typically absorbs at around 600 nm. It is allowed to react with ammonia to form a new complex $[M(NH_3)_6]^{2+}$ that should have absorption at :

(A) 800 nm

☒ (B) 580 nm

(C) 620 nm

☒ (D) 320 nm



$$\Delta_o = \frac{hc}{\lambda}$$

Ans: (B)

The inner orbital complex which exhibits both geometrical as well as optical isomerism.

- (A) $[\text{Cr}(\text{en})_3]^{3+}$ (B) $[\text{IrF}_3(\text{H}_2\text{O})_2(\text{NH}_3)]$ (C) $[\text{NiCl}_2(\text{en})^2]$ (D) $[\text{Co}(\text{CN})_2(\text{ox})_2]^{3-}$

(A) $[\text{Cr}(\text{en})_3]^{3+}$
 Cr^{3+} , d^3
 d^2sp^3 ✓
 geom → X
 optical → ✓
 $\text{en} \rightarrow \text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$
 Symmetrical bidentate

(B) $[\text{IrF}_3(\text{H}_2\text{O})_2(\text{NH}_3)]$
 Ir^{3+} , d^6
 d^2sp^3 ✓
 geom → ✓
 optical → X

(C) $[\text{NiCl}_2(\text{en})^2]$
 Ni^{2+} , d^8
 d^2sp^3 ✓
 geom → ✓
 optical → X

(D) $[\text{Co}(\text{CN})_2(\text{ox})_2]^{3-}$
 Co^{3+} , d^6
 d^2sp^3 ✓
 geom ✓
 optical ✓
 $x - 2 - 4 = -3$
 $x = +3$
 $3d^6$
 SFL

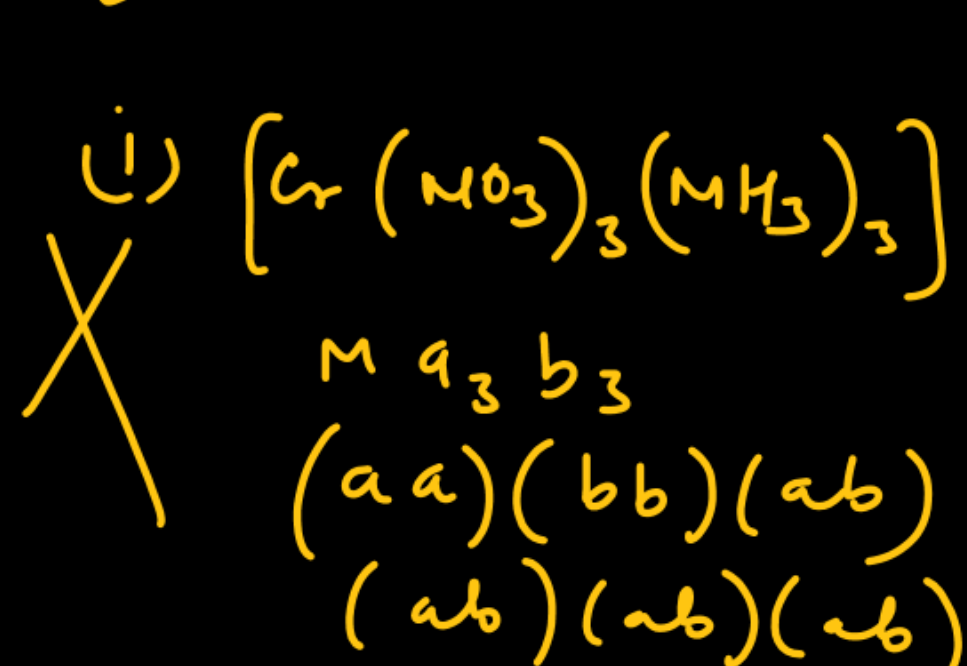
Analysis of (D):
 $(\text{AA})(\text{AA})(\text{bb}) \times$
 $(\text{AA})(\text{Ab})(\text{Ab}) \checkmark$
 sp^3d^2 X
 geom → ✓
 optical → ✓

Final Answer: (D)
 V. Good

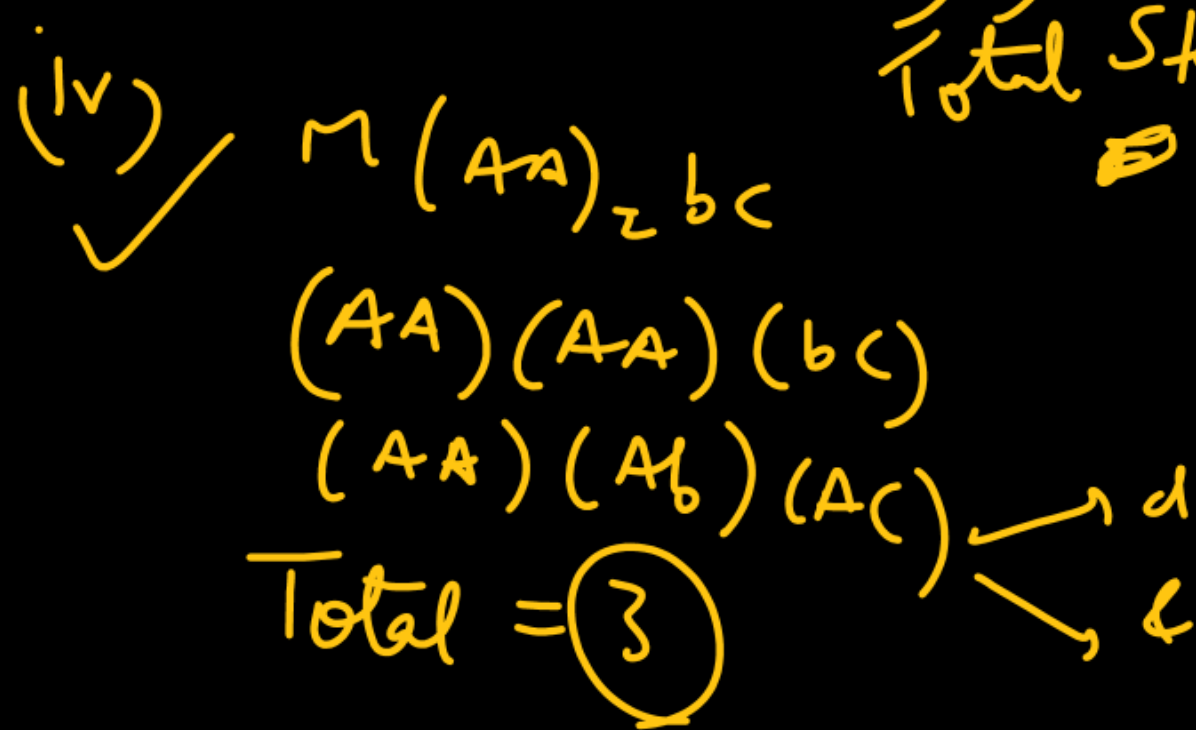
Which of the following will have three stereoisomeric forms?

(i) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (ii) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$ (iii) $\text{K}_3[\text{CoCl}_2(\text{C}_2\text{O}_4)_2]$ (iv) $[\text{CoBrCl}(\text{en})_2]$

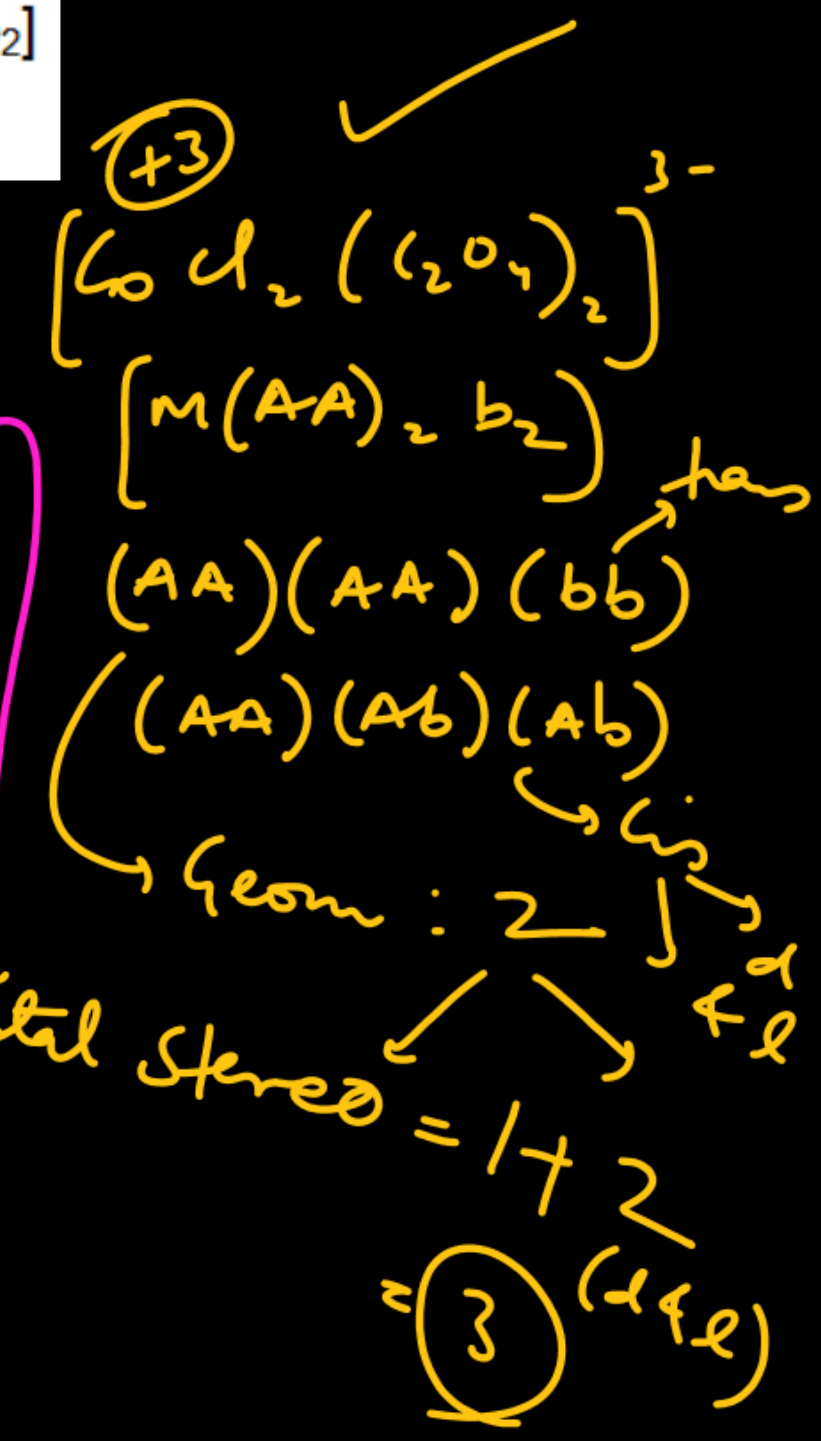
☒ (A) (iii) and (iv) (B) (i), (iii) and (iv) (C) (iv) only (D) All four



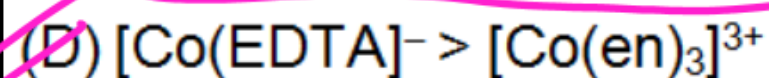
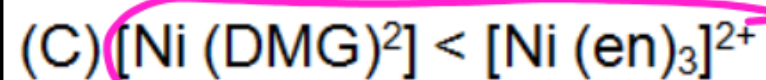
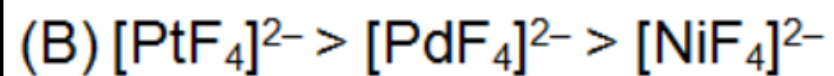
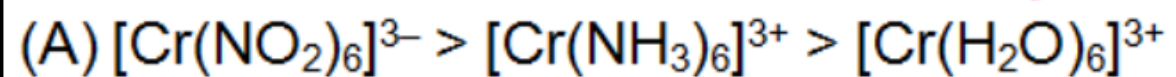
geom — X
 optical — ✓ (2)
 Total Stereo = 2



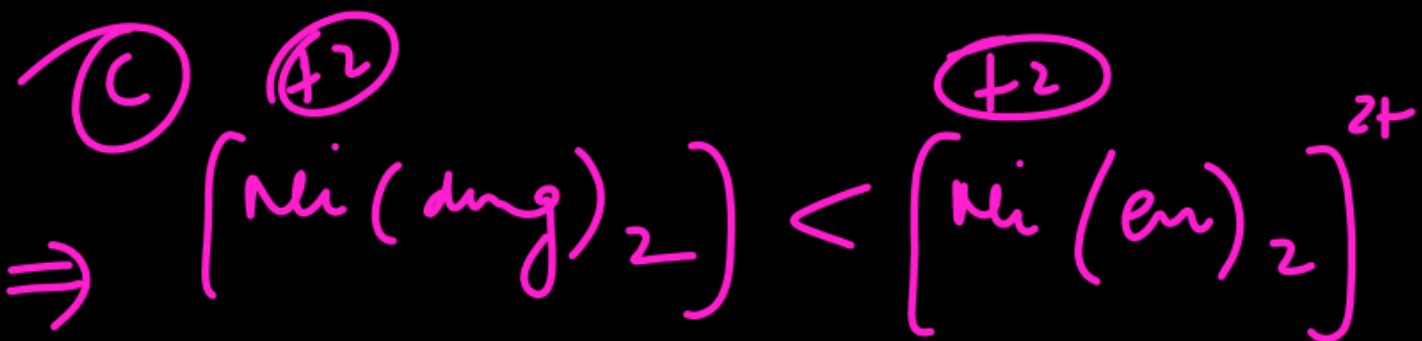
Ans: 4



Which of the following order of CFSE is incorrect?



Correction $\Rightarrow$



$\text{en} > \text{dmg}$
↑
stronger field ligand



↑
stronger field ligand

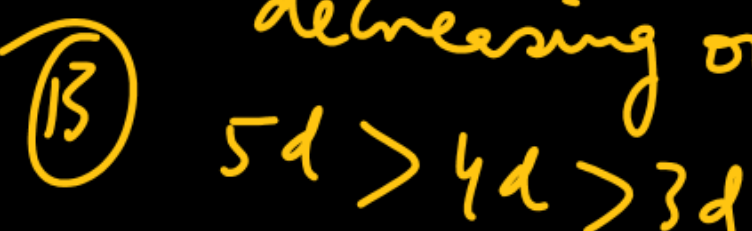
(A) For octahedral complexes,

$$\text{CFSE} = -0.4 \Delta_o \times n_{t_2g} + 0.6 \Delta_o \times n_{e_g} + x \times p$$

$\Delta_o \uparrow, \text{CFSE} \uparrow$



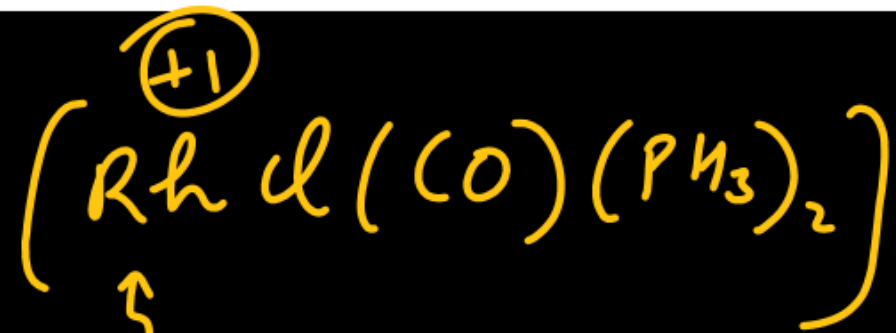
decreasing order of Δ_o .



Ans: (D)

Amongst the following complexes which has square planar geometry?

- (A) $[\text{RhCl}(\text{CO})(\text{PH}_3)_2]$ (B) $\text{K}_2[\text{Cu}(\text{SCN})_4]$ (C) $\text{K}_2[\text{Ni}(\text{PH}_3)_2\text{Cl}_2]$ (D) MnO_4^-

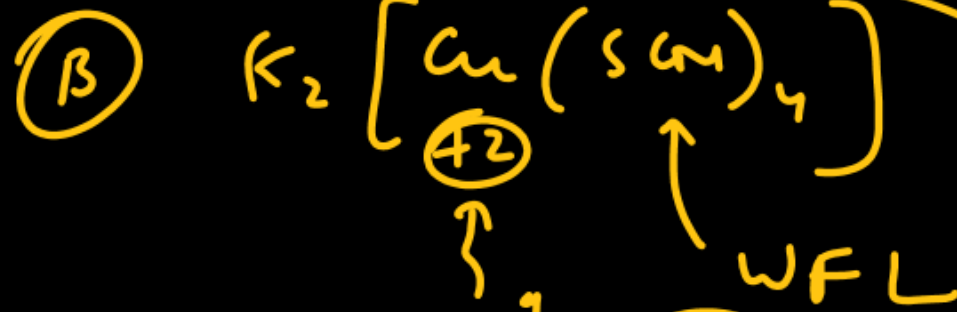


↑

$4d^8 \text{ conf}^n$

(dsp^2)

sq. planar



↑

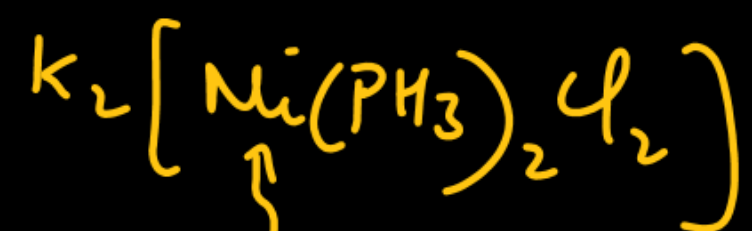
d^9

(sp^3)

Tetrahedral

(C)

$$+2 + x - 2 = 0 \\ x = 0$$



↑

Ni: $3d^8 4s^2$ d^0
: $3d^{10} 4s^0$

(sp^3)

Tetrahedral



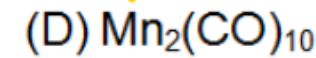
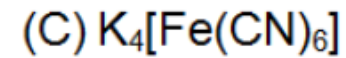
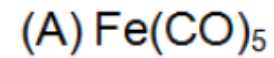
↑

$d^3 s$ (Tetrahedral)

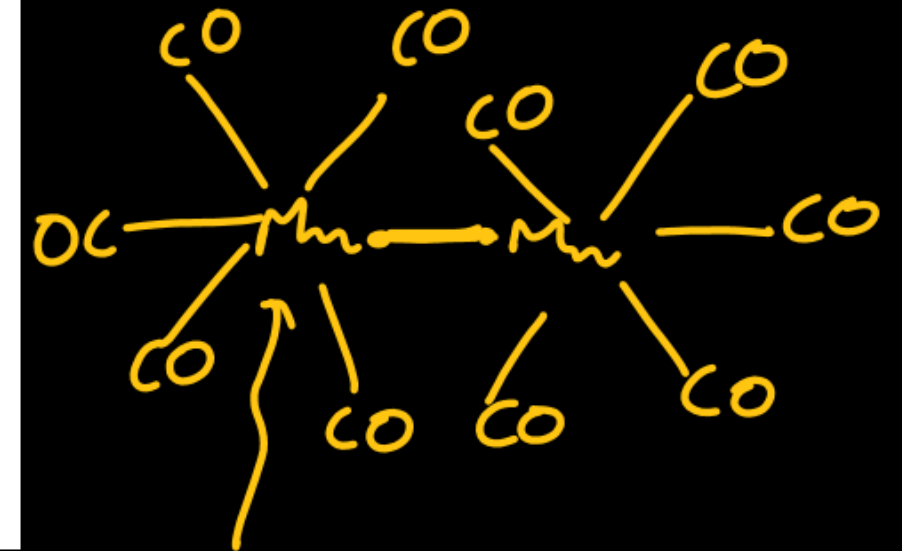
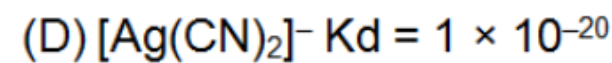
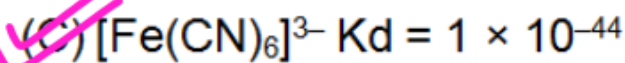
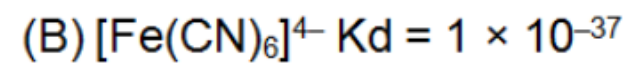
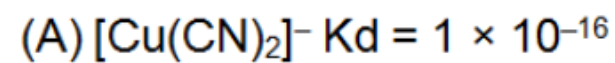
Ans: (A)

On the basis of stability of complex ion in the solution, complexes may be of two type, perfect and imperfect complexes. The stability depends upon the extent of dissociation which in turn depends upon the strength of metal-ligand bond. The stability of complex also depends upon charge on central metal atom, basic nature of ligand, chelation.

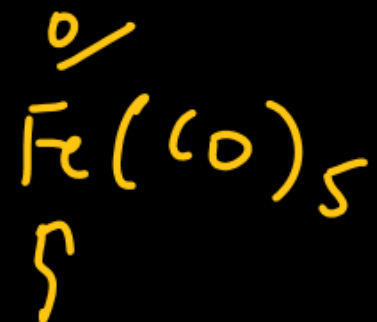
❑ Which one of the following does not follow EAN rule?



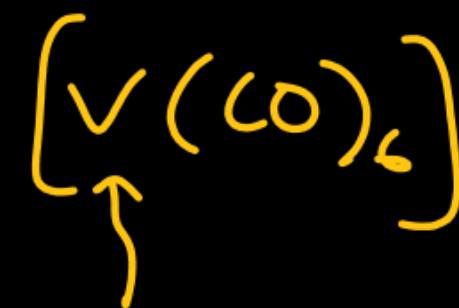
❑ Which complex is most stable?



$$\text{EAN} = 25 - 0 + 10 + 1 = 36$$



EAN rule is same as $18e^-$ rule



$$\text{EAN} = 26 - 0 + 5 \times 2 = 36$$

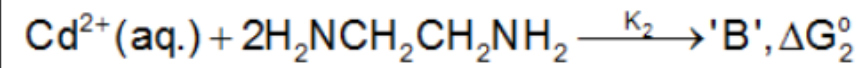
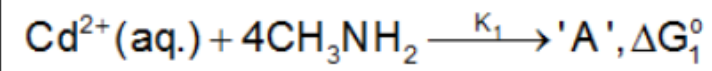
(Follows EAN rule)

$$K_d = \frac{1}{K_f}$$

$$23 - 0 + 12 = 35$$

does not follow EAN. $K_f \uparrow$, stability $\uparrow$. $K_d \downarrow$, stability $\uparrow$.

Consider the following two reactions :



According to given information the correct statement(s) is/are :

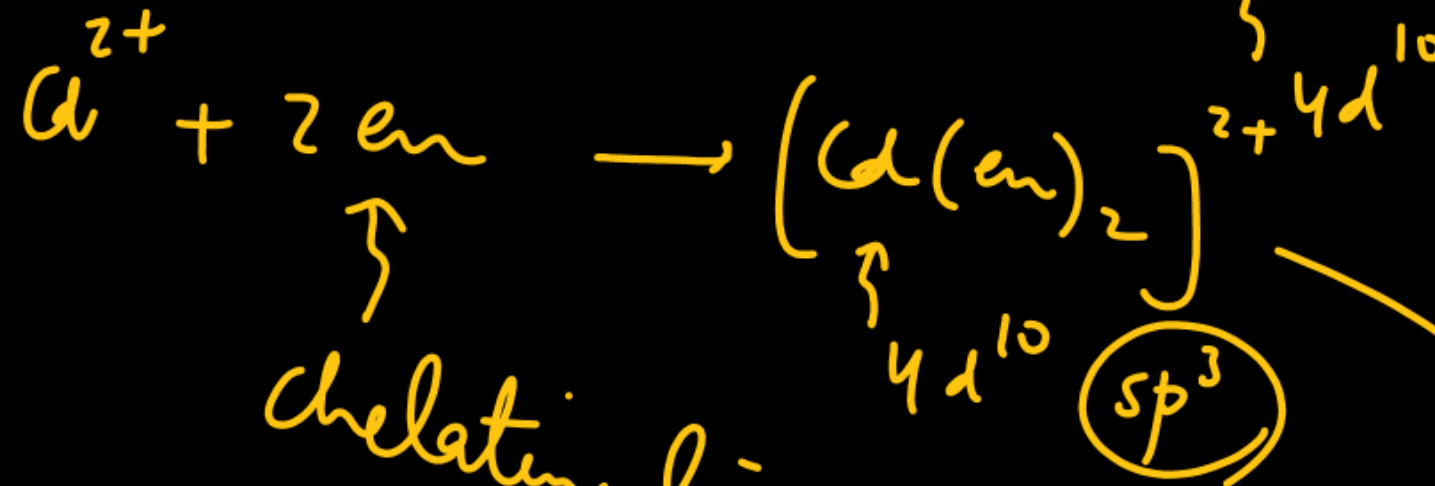
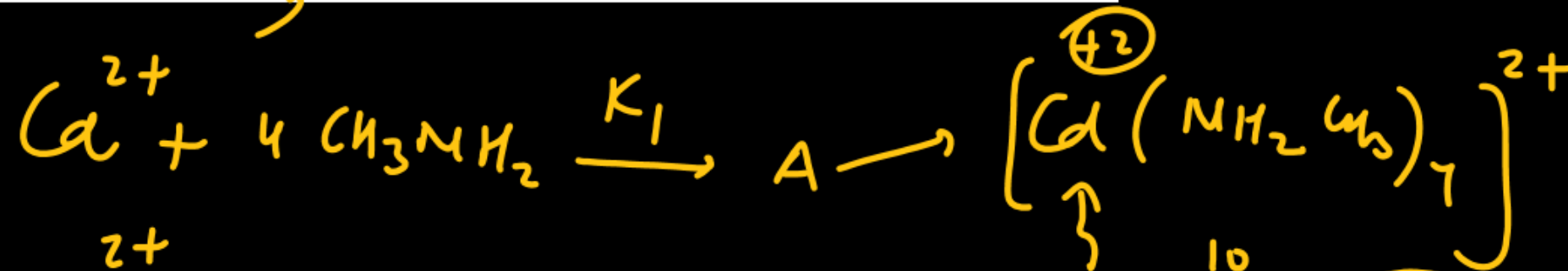
(A) ΔG_2^0 is more negative than ΔG_1^0

(B) Compound 'A' is optically inactive

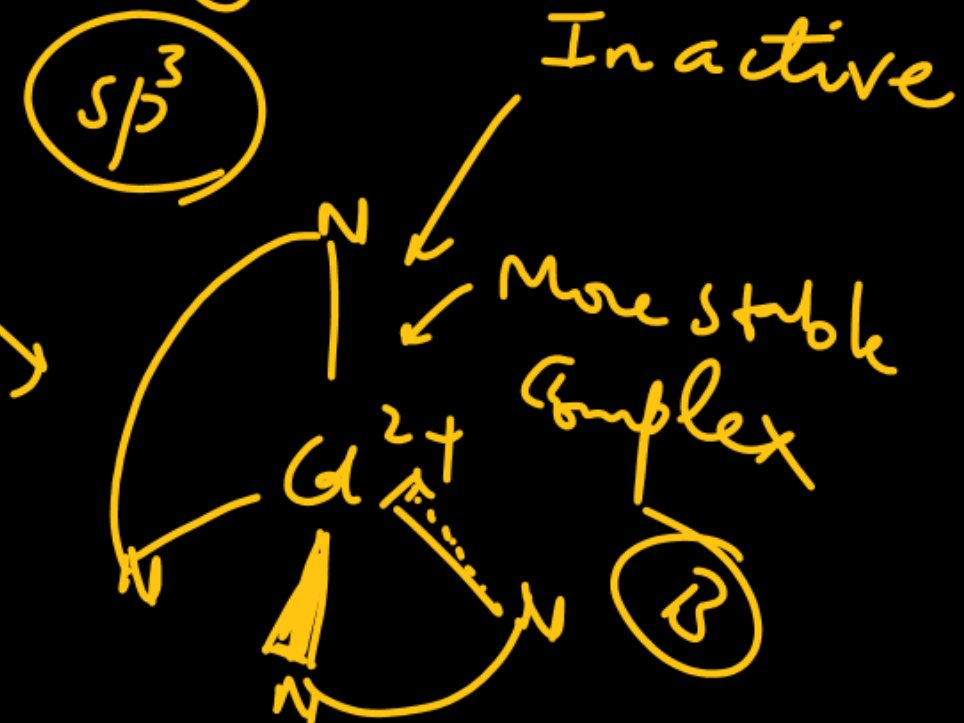
(C) Compound 'B' is optically active

(D) Compound 'AB' is optically active

Ans: A, B



chelating ligand.



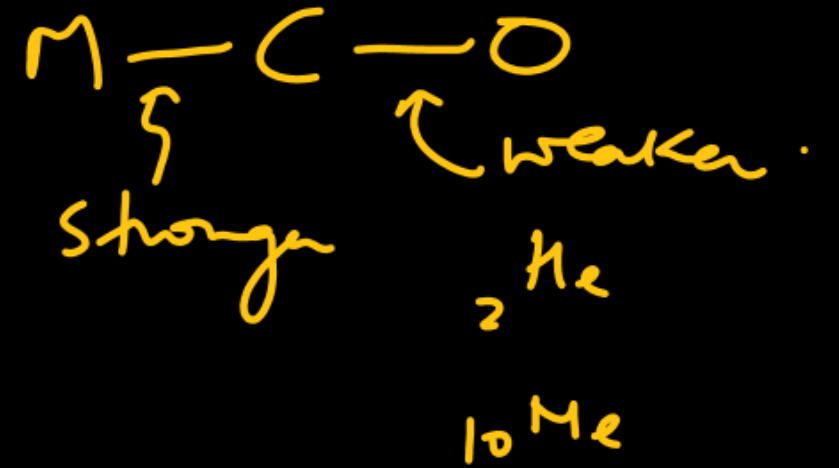
Consider the following two carbonyl compounds

(i) $[\text{Tc}(\text{CO})_6]^+$ and (ii) $[\text{Nb}(\text{CO})_6]^-$

Select incorrect statement (s) for given carbonyl compounds. [Atomic no. Nb = 41, Tc = 43]

- X (A) $[\text{Tc}(\text{CO})_6]^+$ acts as reducing agent and $[\text{Nb}(\text{CO})_6]^-$ acts as oxidizing agent
 X (B) $[\text{Nb}(\text{CO})_6]^-$ acts as reducing agent and $[\text{Tc}(\text{CO})_6]^+$ acts as oxidizing agent
 ✓ (C) "M-C" bond order in $[\text{Nb}(\text{CO})_6]^-$ greater than in $[\text{Tc}(\text{CO})_6]^+$
 X (D) "CO" bond order is greater in $[\text{Nb}(\text{CO})_6]^-$ than in $[\text{Tc}(\text{CO})_6]^+$

Ans : (C)



Synergic bonding

$\text{M}^- > \text{M}^+$

Stronger M-CO bond or synergic bond

Calculation of Effective Atomic Number (EAN) for $[\text{Tc}(\text{CO})_6]^+$:

$$\begin{aligned} \text{EAN} &= 43 - 1 + 12 \\ &= 42 + 12 \\ &= 54 \end{aligned}$$

($Z_{\text{Xe}} = 54$)

54 follows EAN

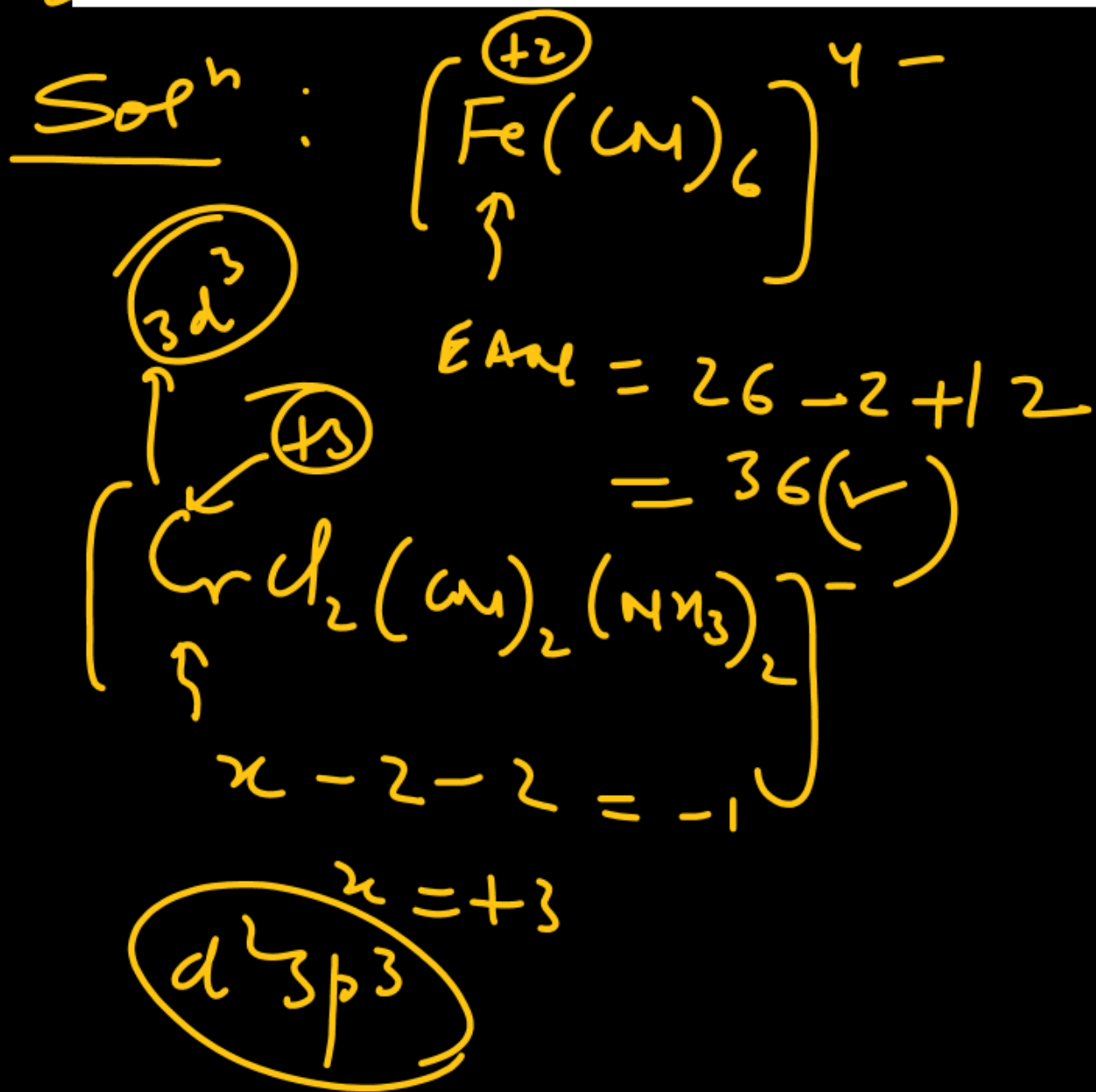
Calculation of Effective Atomic Number (EAN) for $[\text{Nb}(\text{CO})_6]^-$:

$$\begin{aligned} \text{EAN} &= 41 - (-1) + 12 \\ &= 54 \end{aligned}$$

($\underline{\underline{54}}$)

Which of the following statements is/are true?

- ✓ (A) In ferrocyanide ion, the effective atomic number is 36
- ✓ (B) Chelating ligands are atleast bidentate ligand
- ✓ (C) $[\text{CrCl}_2(\text{CN})_2(\text{NH}_3)_2]^-$ and $[\text{CrCl}_3(\text{NH}_3)_3]$ both have d^2sp^3 hybridisation
- ✓ (D) As the number of rings in complex increases, stability of complex (chelate) also increases

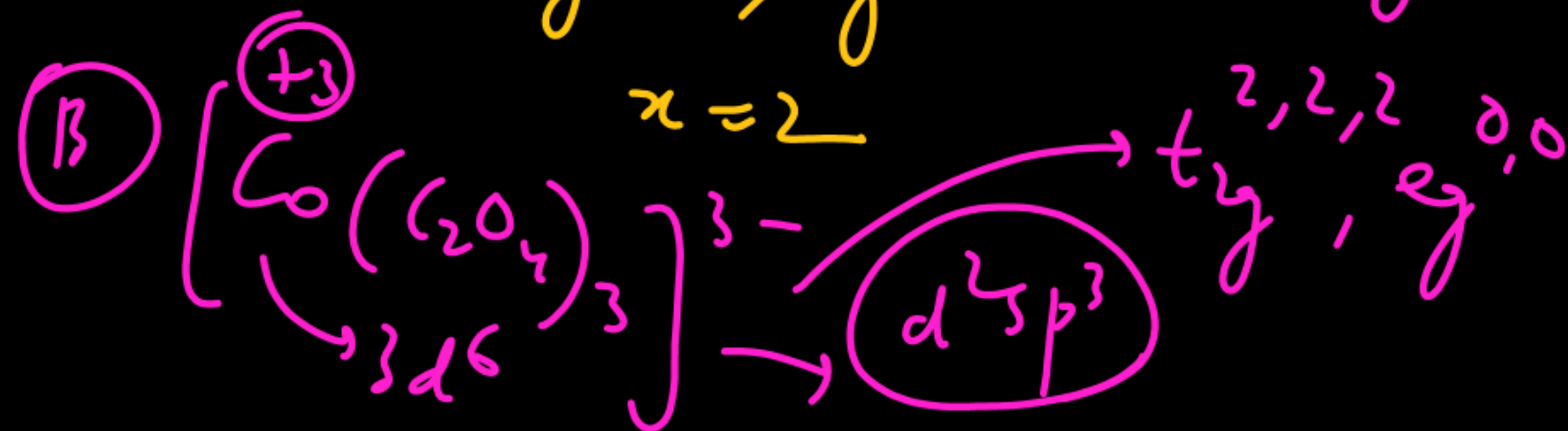
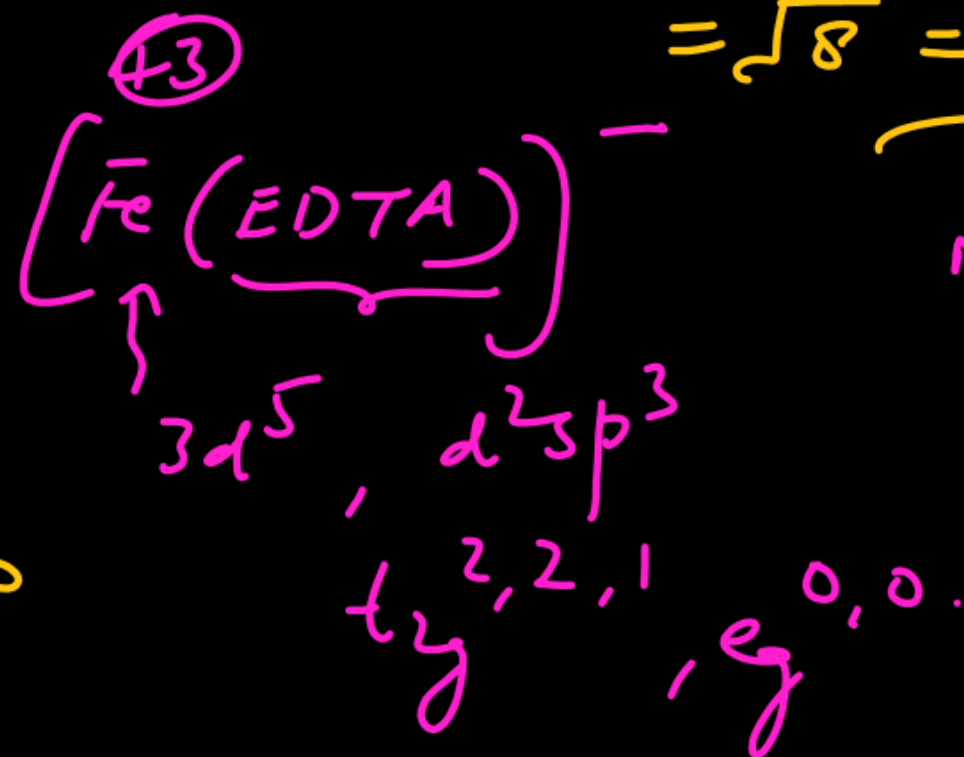
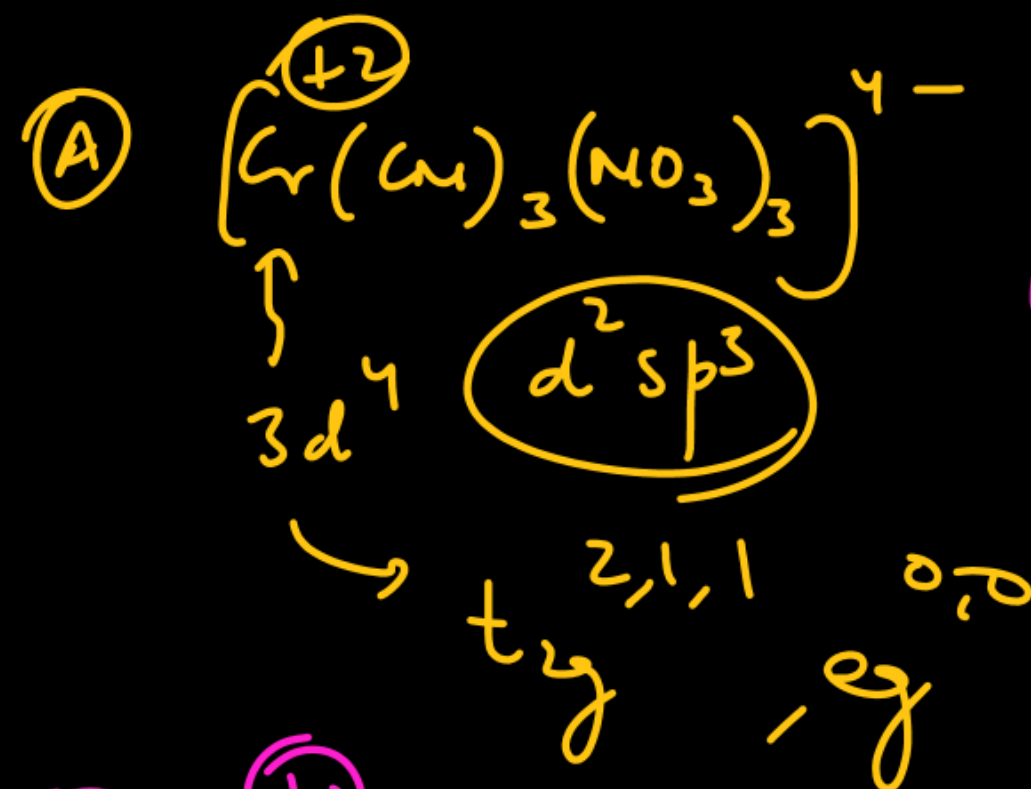
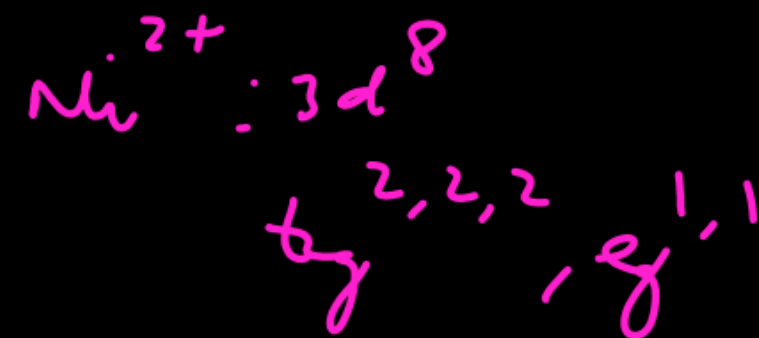


Column-I	Column-II
(A) $[\text{Cr}(\text{CN})_3(\text{NO}_2)_3]^{4-}$	(p) Outer orbital complex
(B) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$	(q) Inner orbital complex
(C) $[\text{Fe}(\text{EDTA})]^-$	(r) $\mu = 2.83 \text{ BM}$
(D) $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$	(s) Shows optical activity
	(t) $\mu = 0$

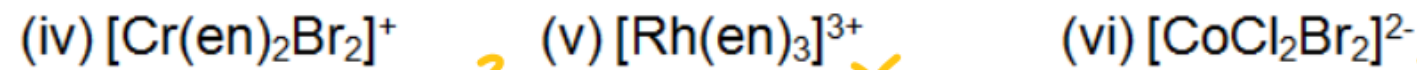
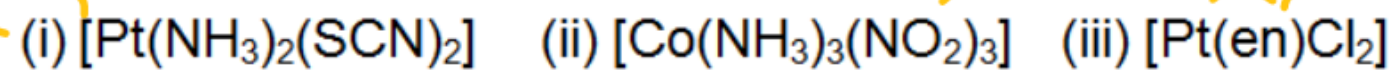
$$\boxed{x = 2}$$

$$\mu = \sqrt{x(x+2)} \text{ B.M.}$$

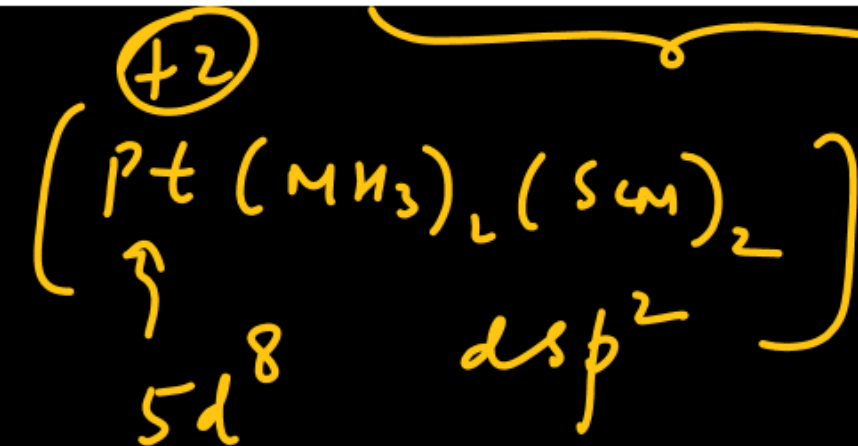
$$= \sqrt{8} = 2.83$$



Consider the following complex compounds :



Then calculate sum of total number of geometrical isomers in all above complex compounds.

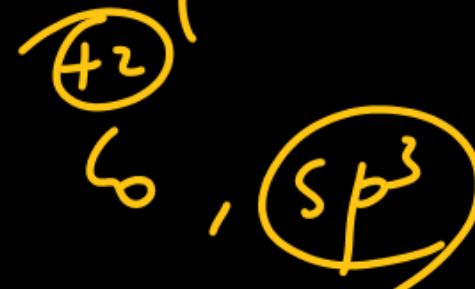


MA_2B_2

$(aa)(bb)$

$(ab)(ab)$

$\rightarrow 2$
geometrical

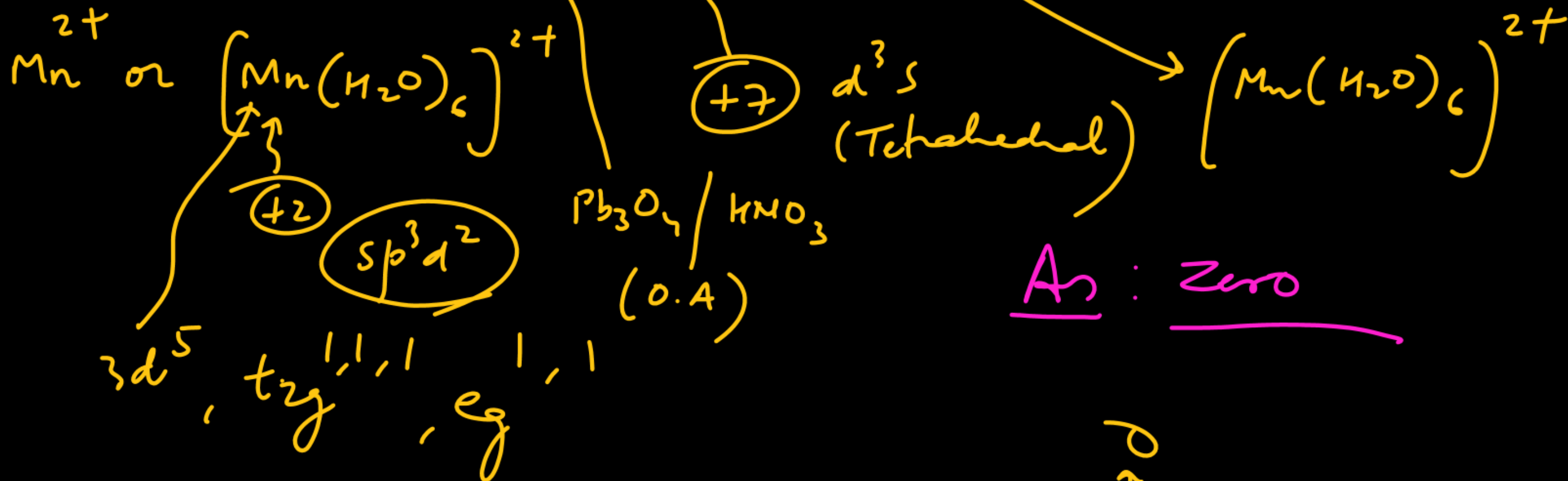


$M(AA)_2B_2 \rightarrow$ No geometrical isomerism.
 $(AA)(bb)$

Ans: 6

A (Light pink colour complex) $\xrightarrow[\Delta]{\text{Pb}_3\text{O}_4 / \text{dil. HNO}_3}$ $\text{HMnO}_4 \xrightarrow{\text{H}_2\text{S} / \text{H}^+}$ A (Light pink colour complex).

Calculate CFSE value in light pink colour complex.



Ans: Zero

$$\text{CFSE} = -0.4 \Delta_o \times n_{t_{2g}} + 0.6 \Delta_o \times n_{e_g} + n \times p \rightarrow 0$$

$$= \text{Zero.}$$